

Crane

Heavy loads are given a **LIFT** by an invention that's happy to do all the hard work

Early cranes

Invented in **ANCIENT GREECE** around **515 BCE**, the crane was adopted and developed further by the Romans. Early Roman cranes used **treadmills powered by slaves**. These machines made it easier to handle and lift building materials and other **heavy items**.

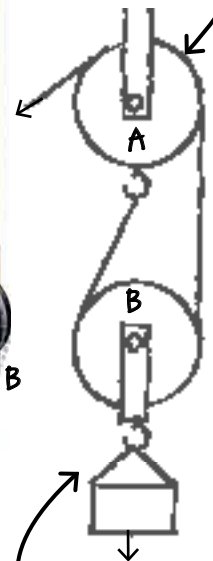
Slaves walking inside the treadmill turn the wheel, which pulls on the rope to lift the load.



Two pulleys connected by a rope are known as a compound pulley.

The pulley (a rope wrapped around a wheel with a groove in it) is the key to a crane's lifting power.

Rope loops over the top pulley, so the rope is pulled down but the load is lifted vertically.



A compound pulley shares the weight of the load between two pulleys, so less lifting force is required.

How it changed
Cranes have taken the weight off workers' shoulders, carrying heavy loads and helping to construct tall buildings that redefine city skylines.
the world

Modern cranes

Today's cranes are **crafted from steel** and powered by engines or electric motors. Tower cranes are used by construction workers to lift the **HEAVY LOADS** involved in erecting tall buildings. Other cranes move **containers** on board ships or carry objects around factories.

